

Helix OTA — Cuttlefish Rootful-Privileged Container: §11.4.161 Documented Exception

Field	Value
Revision	1
Last modified	2026-06-22T00:00:00Z
Status	documented-exception recorded — narrowest-scope rootful-privileged Cuttlefish launch; image build + artifact fetch remain rootless The §11.4.161 (rootless-container-runtime mandate) documented exception for the Cuttlefish (cvd) virtual-Android Tier-2 OTA validation path. Rootless Podman is NOT viable for Cuttlefish: the official google/android-cuttlefish container README mandates --privileged (kvm + vsock + network), and a rootless container cannot create the cvd-ubr bridge / cvd-wtap/cvd-mtap tap devices nor write the /proc/sys/net knobs Cuttlefish needs (these require CAP_NET_ADMIN in a root network namespace; rootless uses slirp4netns/pasta). This is classified per §11.4.112 as a host-gated documented exception, NOT a structural impossibility — Cuttlefish IS possible, it merely needs container privilege. Every load-bearing claim is sourced (see ## Sources verified 2026-06-22); uncertain claims are marked UNCONFIRMED: per §11.4.6.
Status summary	
Authority	Helix OTA control-plane / device-integration team docs/design/CUTTLEFISH_TIER2.md (the Tier-2 bring-up recipe this exception authorises); docs/design/EMULATED_DEVICE_TESTING.md (Tier-1/2/3 overview); containers/submodule (vasic-digital/
Related	

Field	Value
	containers, §11.4.76 / §11.4.161) — proposed pkg/cuttlefish extension

1. Purpose + scope of this exception

§11.4.161 (rootless container runtime mandate) requires every containerized workload to run under **Podman in rootless mode**; Docker-rootful, sudo, or any escalation to root for container management is **FORBIDDEN unless the target platform has no rootless option AND that constraint is documented per §11.4.112**.

This document IS that documented constraint, for exactly one workload: the **launch of the Cuttlefish (cvd) virtual Android device inside a container** for Tier-2 OTA validation (real update_engine + AVB/dm-verity + auto-rollback — the path docs/design/CUTTLEFISH_TIER2.md details). It establishes as cited FACT that rootless Podman cannot host Cuttlefish, records the minimal-privilege rootful recipe actually used, the residual risk + reversibility, and the strict scope-narrowing (only the launch_cvd run is privileged; the container-image build and the AOSP artifact fetch stay rootless).

2. Classification (§11.4.112): documented exception, NOT structural-impossibility

Per §11.4.112 the correct classification is a **host-gated / documented exception**, NOT structurally-impossible won't-fix:

- **Cuttlefish IS possible** — it boots and applies real OTAs on a Linux + KVM host. The only requirement that conflicts with §11.4.161 is **container privilege** (`--privileged`
 - the device passthrough + host networking). This is a *capability requirement*, not a platform/protocol impossibility.
- Contrast with a true §11.4.112 structural-impossibility case (e.g. relocating a FLAG_SECURE secure surface to a second display, which the platform blanks by design): there the goal cannot be achieved on the platform at all. Here the goal IS achievable — it simply requires the privilege the rootless mandate normally forbids, so it is recorded as a **§11.4.161 documented exception** rather than a won't-fix.
- A future change MAY narrow or remove the exception (see §6) — the classification is durable but not eternal; a reopen needs NEW evidence rootless Cuttlefish became viable (§11.4.7 / §11.4.34), never a re-derivation of the same constraint.

3. Why rootless Podman cannot host Cuttlefish (cited FACT)

The official Cuttlefish container path is privileged BY DESIGN:

- The google/android-cuttlefish container/README.md documents the canonical run as a **--privileged** container with KVM, vsock, and host networking — Cuttlefish inside a container is run with **--privileged** and the host's /dev/kvm, /dev/vhost-vsock, /dev/vhost-net, /dev/vsock, and /dev/net/tun passed through [SRC-CF-CONTAINER].
- **Networking needs a root network namespace.** Cuttlefish creates a bridge (cvd-ubr) and per-instance tap devices (cvd-wtap-NN / cvd-mtap-NN) and writes /proc/sys/net/* knobs to wire guest networking; these operations require **CAP_NET_ADMIN in a root (initial) network namespace** [SRC-CF-GET-STARTED] [SRC-CF-ONPREM]. A **rootless** container has no such namespace — rootless Podman provides userspace networking via slirp4netns/pasta, which **cannot** create host bridges/tap devices nor write the /proc/sys/net knobs [SRC-PODMAN-PRIV] [SRC-ROOTLESS-NET] [SRC-NONPRIV-POD]. The bridge/tap creation therefore fails under rootless, and Cuttlefish guest networking does not come up.
- **Device + KVM access.** /dev/kvm + the vhost_vsock/vhost_net device nodes must be accessible to the guest VMM; the documented path grants this via **--privileged** device passthrough rather than ad-hoc per-device rootless mapping [SRC-CF-CONTAINER].

Net FACT: the project's deep research (§11.4.150, accessed 2026-06-22) found **no** rootless Podman recipe that brings up Cuttlefish networking — the upstream-mandated **--privileged** path is the only documented, working one. Per §11.4.161 this is the “no rootless option” condition, and this document is the required §11.4.112 record.

4. The minimal-privilege rootful recipe actually used (FACT)

Only the **launch step** is privileged; everything narrower is preferred where it works.

```
# Rootful Podman, narrowest device + network grant that Cuttlefish
# requires.
# (run as the privileged launch step ONLY – see §5 scope
# narrowing)
podman run --rm \
  --privileged \
  --network host \
  --device /dev/kvm \
  --device /dev/vhost-vsock \
  --device /dev/vhost-net \
  --device /dev/vsock \
  --device /dev/net/tun \
  <cuttlefish-image> \
  /bin/bash -lc 'HOME=$PWD ./bin/launch_cvd --daemon && ./bin/adb
  devices'
```

Host preconditions (FACT):

- **Kernel modules** vhost_vsock + vhost_net loaded on the host (or built-in). On the current candidate host nezha, lsmod does **not** list them, which is consistent with them being **built-in** to the host's Linux 6.12 kernel — **UNCONFIRMED**: whether built-in vs not-yet-loaded; verify at bring-up with modinfo vhost_vsock / checking /sys/module/vhost_vsock before the first privileged launch (do not guess).
- **Host groups** kvm, cvdnetwork, render for the launching user [SRC-CF-GET-STARTED].
- **Device nodes.** On nezha (FACT, host probe 2026-06-22): /dev/kvm, /dev/vhost-vsock, /dev/vhost-net, /dev/net/tun are present and world-rw; /dev/vsock is **ABSENT**. The --device /dev/vsock line above is therefore conditional — include it only if the node exists at launch time; **UNCONFIRMED**: whether Cuttlefish on this kernel needs the /dev/vsock client node in addition to /dev/vhost-vsock, verify on first bring-up rather than assume.
- **Guest target.** Tier-2 fetches the public-CI target **aosp_cf_x86_64_only_phone-userdebug** from ci.android.com — **no credentials required** [SRC-CF-GET-STARTED]. The update_engine README confirms “Cuttlefish works as well” for the real A/B update_engine apply [SRC-UE-SEARCH].

5. Scope narrowing — only the launch is privileged

The exception is the **narrowest** possible. The Tier-2 pipeline has three phases; only the third is privileged:

1. **Container-image build — ROOTLESS.** Building the Cuttlefish container image (Debian host packages per docs/design/CUTTLEFISH_TIER2.md §4.1) is an ordinary rootless Podman build — no privilege, no exception.
2. **AOSP artifact fetch — ROOTLESS, no credentials.** Downloading the device-image zip + cvd-host_package.tar.gz for aosp_cf_x86_64_only_phone-userdebug from the public ci.android.com is a plain rootless network fetch — no privilege, no secrets [SRC-CF-GET-STARTED].
3. **launch_cvd run — ROOTFUL --privileged (THIS exception).** Only the actual boot of the virtual device (which creates the cvd-ebr bridge + tap devices and opens /dev/kvm + vhost nodes) requires the privileged rootful container of §4.

Everything that CAN be rootless STAYS rootless; the privileged surface is confined to the single launch_cvd/stop_cvd lifecycle. This is the §11.4.161 “documented + narrowest” posture.

6. Residual risk + reversibility

- **Residual risk.** A --privileged --network host container has broad host access for the duration of the launch (device nodes + host network namespace). Mitigations: it runs **only** the launch_cvd/stop_cvd lifecycle on a dedicated Linux test host (nezha), not on the operator's primary workstation; it carries no project secrets (§11.4.10 — the AOSP fetch is credential-free); and the

privileged window is the launch only (§5), torn down with `stop_cvd + container removal`.

- **Reversibility (FACT).** The exception leaves **no persistent host mutation beyond reversible config**: the Cuttlefish host packages are apt-installed (removable via `apt remove cuttlefish-base cuttlefish-user`), the only host-state changes are the loaded kernel modules (`vhost_vsock/vhost_net`, transient) and the user's group membership (`kvm/cvdnetwork/render`, reversible via `gpasswd -d`). No bootloader, partition, or firmware change occurs on the host — the A/B slot-switch + corrupt-slot-rollback happen **inside the virtual device**, never on the host (composes §11.4.133 target-safety: the “target” is the Cuttlefish guest, not the host).
- **Containers-submodule routing (§11.4.76 / §11.4.161).** The privileged launch **MUST** be driven through the `vasic-digital/containers` submodule's lifecycle primitives — the proposed `pkg/cuttlefish` extension wrapping `launch_cvd/stop_cvd + adb devices` readiness — NOT ad-hoc `podman run` outside `pkg/boot/pkg/compose/pkg/health`. The `--privileged` flag is requested through the submodule's run options, keeping the exception inside the sanctioned orchestration layer; a missing capability is extended upstream per §11.4.74, never worked around in-project.

7. Honest boundary (§11.4.6)

- This document authorises the **container privilege** Cuttlefish needs; it does NOT claim Tier-2 has run. The Tier-2 bring-up + apply recipe and its still-open UNCONFIRMED: items (exact Virtual-A/B-vs-legacy on the chosen image; the exact corrupt-slot mechanism) live in `docs/design/CUTTLEFISH_TIER2.md` and remain open there.
- The exception is **operator-gated for first use**: the first privileged `launch_cvd` requires operator authorisation to run a rootful `--privileged` container on the test host (high-blast-radius capability grant, §11.4.101 / §11.4.21). Until then the path is designed + authorised-in-doc, pending operator privileged-launch authorization — not a green, not Operator-blocked beyond that one authorisation.

Sources verified 2026-06-22

- [SRC-CF-CONTAINER] `google/android-cuttlefish` — `container/README.md`, GitHub — <https://github.com/google/android-cuttlefish> (Cuttlefish-in-a-container run uses `--privileged` with host `/dev/kvm`, `/dev/vhost-vsock`, `/dev/vhost-net`, `/dev/vsock`, `/dev/net/tun` passthrough + host networking) — accessed 2026-06-22.
- [SRC-CF-GET-STARTED] Get started — Cuttlefish, Android Open Source Project — <https://source.android.com/docs/devices/cuttlefish/get-started> (host KVM + group membership `kvm/cvdnetwork/render`; `ci.android.com` image fetch; public `aosp_cf_x86_64_only_phone-userdebug` target, no credentials) — accessed 2026-06-22.
- [SRC-CF-ONPREM] On-premises Cuttlefish, AOSP — <https://source.android.com/docs/devices/cuttlefish/on-premises> (host networking / `cvd-ebr` bridge + tap-device requirements for multi-instance Cuttlefish networking) — accessed 2026-06-22.

- [SRC-PODMAN-PRIV] Podman run --privileged docs — privileged containers grant the device + capability access that rootless containers lack; rootless cannot create host bridges/tap devices — accessed 2026-06-22.
- [SRC-ROOTLESS-NET] Rootless container networking (oneuptime / Podman rootless networking writeup) — rootless uses slirp4netns/pasta userspace networking and cannot perform CAP_NET_ADMIN host-namespace operations (bridge/tap creation, /proc/sys/net writes) — accessed 2026-06-22.
- [SRC-NONPRIV-POD] Non-privileged Cuttlefish-pod community writeup (Medium) — documents why a non-privileged pod cannot create the Cuttlefish bridge/tap network and the privileged path it required instead — accessed 2026-06-22.
- [SRC-UE-SEARCH] platform/system/update_engine README / testing notes, Git at Google — “Cuttlefish works as well” for testing the real update_engine A/B apply — accessed 2026-06-22.

Negative findings / gaps (§11.4.99(B))

- Deep research (§11.4.150) surfaced **no** working rootless-Podman Cuttlefish recipe; every documented, working path is --privileged. Absence of a rootless recipe across the official container README + AOSP on-premises docs + community writeups is the cited basis for “no rootless option” under §11.4.161 — not an assumption.
- **UNCONFIRMED:** whether nezha’s vhost_vsock/vhost_net are built-in vs unloaded, and whether the /dev/vsock client node is required in addition to /dev/vhost-vsock on this kernel — both are verify-before-first-launch items (§4), not blockers to recording this exception.